AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY



Department of Computer Science and Engineering

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Course Title: Data Structures Lab

Assignment No: 06

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Question:

Write a program to support multi-digit numbers in the conversion of an infix expression to postfix notation and the evaluation of the resulting postfix expression. For example, your code should convert this infix expression to its appropriate postfix notation: (12 + 34) * 56 - 78 / 9. Then evaluate the result of this expression from the postfix notation.

Solution Code:

```
#include<bits/stdc++.h>
using namespace std;

int priority(char s) {
   if(s == '^') return 100;
   else if(s == '*' || s == '/') return 50;
   else if(s == '+' || s == '-') return 20;
   else return 0;
}

int main() {
   string infix, postfix = "";
   getline(cin, infix);

   stack<char> s;
   for(int i = 0; i < infix.size(); i++) {

       if(infix[i] == ' ')
           continue;

       char symbol = infix[i];


       if(isdigit(symbol)) {
           while(i < infix.size() && isdigit(infix[i])) {
               postfix += infix[i];
               i++;
           }
           postfix += ' ';
           i--;
       }

       else if(symbol == '(') {
           s.push(symbol);
       }
```

```
    else if(symbol == ')') {
        while(!s.empty() && s.top() != '(') {
            postfix += s.top();
            postfix += ' ';
            s.pop();
        }
        s.pop();
    }

    else {
        while(!s.empty() && priority(s.top()) >= priority(symbol)) {
            postfix += s.top();
            postfix += ' ';
            s.pop();
        }
        s.push(symbol);
    }
}

while(!s.empty()) {
    postfix += s.top();
    postfix += ' ';
    s.pop();
}

cout << "The postfix expression is: " << postfix << "\n";

stack<int> s2;

stringstream ss(postfix);
string token;

while(ss >> token) {
    if(isdigit(token[0])) {
        s2.push(stoi(token));
    }
    else {
        int a = s2.top(); s2.pop();
        int b = s2.top(); s2.pop();

        int c;

        if(token == "+")       c = b + a;
        else if(token == "-")  c = b - a;
        else if(token == "*")  c = b * a;
        else if(token == "/")  c = b / a;
        else if(token == "^")  c = pow(b, a);

        s2.push(c);
    }
}
```

```
    cout << s2.top() << endl;

    return 0;
}
```

Sample Input Output:

```
(12 + 34) * 56 - 78 / 9
The postfix expression is: 12 34 + 56 * 78 9 / -
2568
```